

Selection (server, auto) → Debate (in-session, LLM) → Persist + Surface

DATA
LAYER

STEP 1: DATA REFRESH

price_discovery.py -> .scan_cache.pkl (770 ticker x 80+ fields)
uvicorn /api/reload -> api.py STATE

SELECTION
(server-side
auto)

STEP 2: SELECTION (auto, no LLM)

BuyScore = Composite + Class +/- OER + Consensus + 1M/Sector regime - WeakQVR + ETF flag
Top 5 stock + Top 5 ETF DESC (LONG) / Bottom 5 stock + Bottom 5 ETF ASC (SHORT)
Rotation cooldown (2-week) -> 20 ticker targets

STEP 3: SPECIALIST PROMPT BUILD

Per ticker x 3 personas (Fundamental / Sentiment / Valuation)
Inject quant readings (Composite, OER, TCS, TFS, RSS, URS, QVR)
Lane enforcement: STAY IN YOUR LANE

MULTI-AGENT
DEBATE
(in-session
LLM only)

STEP 4: ROUND 1 - 3 PARALLEL SUB-AGENT CALLS

Agent(market-researcher, fund_prompt) + sent_prompt + val_prompt
3 specialists work independently, no cross-knowledge of others
~30-40s per call (parallel)

STEP 5: PARSE

parse_specialist_opinion(): free-text -> SpecialistOpinion
Extract rating (BUY/HOLD/SELL/AVOID), confidence (0-1), key_points

STEP 6: CONVERGENCE CHECK

CONSENSUS_BUY/HOLD/SELL (1 tier) -> STOP
ENTRY_TIMING (BUY/HOLD) / EXIT_TIMING (HOLD/SELL) (adjacent) -> STOP
POLAR_SPLIT (BUY/SELL) / WIDE_SPAN (3 tiers) -> R2 trigger

STEP 7 (conditional): ROUND 2 CROSS-EXAM

Only if POLAR_SPLIT or WIDE_SPAN. Each specialist sees other 2's R1 opinions.
Re-issue rating + confidence delta. Usually narrows but rating may persist.

SYNTHESIS
+ PERSIST

STEP 8: DUAL SYNTHESIS (rule-based, no LLM)

NEUTRAL mode: confidence-weighted ensemble -> rating + modifier [-5..+5]
AVERSE mode: bear/risk language x1.2 weight -> typically 1 tier more conservative
risk_mode_gap = |neutral_tier - averse_tier|

STEP 9: PERSIST + SURFACE

.multi_agent_debate_cache.json (immutable, append-only)
/api/conviction-debate/multi -> selection (fresh) + cache (lookup) merge
Dashboard 4 sections: Stock LONG/SHORT, ETF LONG/SHORT
Pages 2-5 expand each step + walk through 2 real cases (CNQ consensus / QCLN polar-split).

Each step: what goes in, what happens, what comes out

1. DATA REFRESH

IN: user trigger or cron
DO: scan 770 ticker, compute 80+ fields
OUT: .scan_cache.pkl + api.py STATE

2. SELECTION

IN: scan_cache + sector regime + consensus
DO: BuyScore -> rank DESC/ASC by asset
OUT: Top 5+5 long + Top 5+5 short = 20

3. PROMPT BUILD

IN: 20 targets + quant readings per ticker
DO: formula 3 persona prompts per ticker
OUT: 60 prompts (20x3) total

4. ROUND 1

IN: 60 prompts
DO: 3 parallel sub-agent calls per ticker
OUT: 60 free-text specialist opinions

5. PARSE

IN: 60 free-text
DO: regex + bullet extract
OUT: 60 SpecialistOpinion (rating, conf, ...)

6. CONVERGENCE

IN: per ticker {Fund, Sent, Val} ratings
DO: classify 6 patterns
OUT: converged? type label

7. ROUND 2 (rare)

IN: POLAR_SPLIT/WIDE_SPAN only (~10-30%)
DO: cross-exam: inject other 2 R1
OUT: updated opinions (rating + conf delta)

8. DUAL SYNTHESIS

IN: final round per ticker
DO: rule-based blend (Neutral / Averse)
OUT: 2 verdict per ticker (mod + reasoning)

9. PERSIST + SURFACE

IN: MultiAgentVerdict per ticker
DO: append cache + serve via API
OUT: dashboard renders 4 sections

Flow: 1 -> 2 -> 3 -> 4 -> 5 -> 6 (-> 7 if needed) -> 8 -> 9

Steps 4 + 7 are the ONLY LLM calls (in-session, Claude Max plan). All other steps are pure-Python or server-side cache I/O.

Case A: CNQ — CONSENSUS_BUY (R1 stop)

Fastest path: 3 specialists agree, R2 skipped, full target weight in both modes

Step 1-2: Quant readings + BuyScore selection

Composite 74.9 | Class CONTINUATION | OER 0 (PRISTINE) | TCS 100/85 | RSS 74/74 | Pre-Mom 0.8 STRONG
BuyScore = 95.86 -> Stock LONG Top 1 / 5
Comment: OER 0 at Composite >70 occurs in <3% of universe -> rare 'trend with max room' setup.

Step 3-5: Round 1 Parallel (3 sub-agent calls, ~30s)

FUNDAMENTAL BUY conf 0.80	SENTIMENT BUY conf 0.70	VALUATION BUY conf 0.85
60+ yr reserve life moat TMX structural -\$4 differential lift 100% FCF return policy \$30-35/bbl breakeven among lowest globally DCF +15-20% upside vs current	TMX 12m structural lift recognized Q1 in-line + 7% dividend hike Berkshire 6% stake unchanged (sticky) Carney govt energy-supportive TMX2 rumor as optionality	Composite 74.9 top decile OER 0 PRISTINE (very rare) TCS 100/85 max-strength trend RSS 74/74 sector leadership Distribution days = 0

Step 6: Convergence Check

Result: CONSENSUS_BUY (1 tier, dispersion 0.06)

All 3 specialists rate BUY with confidence 0.70-0.85. rating_axis = 0. Convergence detector returns TRUE.
Round 2 skipped — synthesis proceeds directly. converged_round = 1.

Step 8: Dual Synthesis (rule-based, no LLM)

NEUTRAL mode: BUY +3

Confidence-weighted ensemble: 0.80+0.70+0.85 = 2.35 all-BUY.

Reasoning: Triple BUY consensus with the rarest possible OER 0 + Composite 75.
TMX structural lift + Berkshire anchor + dividend coverage.

SIZING: Full target weight. Add on any 5-8% pullback to SMA50.

AVERSE mode: BUY +3 (same!)

Risk-language weighting does NOT change verdict because:
- OER 0 = zero tactical drawdown risk
- 5.2% dividend yield = downside protection
- Berkshire anchor reduces forced-seller risk

SIZING: Full target weight. Averse-default (gap = 0).

Step 9: Persist + Surface

Cache entry: { ticker: CNQ, side: long, synthesis_neutral: BUY+3, synthesis_averse: BUY+3, converged_round: 1, disagreement: {type: CONSENSUS_BUY, rating_axis: 0, dispersion: 0.06} }
Dashboard: Stock LONG section, row 1, both Neutral and Averse cells green BUY +3, Gap badge green (0), R = 1.

Fund SELL vs Sent/Val BUY -> R2 triggered -> narrows confidences but rating spread persists -> risk-mode dependent verdict

Step 1-2: Quant readings

Composite 68.7 | Class OVEREXTENDED | OER 58 | HEALTHY_TREND breadth 72% | BuyScore 78 -> ETF LONG candidate
Comment: Strong technical + flow signals, but fundamental specialist must evaluate basket quality independently.

Step 3-5: ROUND 1 Parallel — divergent ratings

FUNDAMENTAL SELL conf 0.72	SENTIMENT BUY conf 0.62	VALUATION BUY conf 0.78
ROIC 6.5% < WACC 8% (value-destroying basket) GM 32% vs SPY 41% 22% holdings -EPS IRA dependence 73%	IRA continuity post-election AI-PPA demand floor Inflow \$80M reverses 6m outflow Goldman OW thematic Reddit shift contrarian-long	Composite top 5% HEALTHY_TREND flag TFS>TCS fresh formation Pre-Mom 0.8 STRONG RSS 78/83 leadership

Step 6: Convergence -> POLAR_SPLIT (SELL ^ BUY both present)

rating_axis = 2 (max). rset = {SELL, BUY}. Convergence detector returns FALSE.
Trigger Round 2 cross-examination. converged_round = 0 (will remain if R2 also fails).

Step 7: ROUND 2 — cross-examination (3 more sub-agent calls)

FUND -> SELL 0.62	SENT -> BUY 0.57	VAL -> BUY 0.75
Acknowledges HEALTHY_TREND flag + flow inflection (conf -0.10) Rating UNCHANGED: basket quality concern overrides tape signals	Acknowledges Fund's IRA-dependence + ROIC<WACC tail risks (conf -0.05) Rating UNCHANGED but conviction reduced from 0.62 to 0.57	Acknowledges Fund's basket-quality lane Defers to that lane (conf -0.03) Pure technical view unchanged: HEALTHY_TREND + breadth still valid

R2 result: still POLAR_SPLIT. Conf deltas narrow gap but rating spread persists. STOP at R2. converged_round = 0 (signals 'no full consensus').

Step 8: Dual Synthesis

NEUTRAL: BUY +1 (mild) Conf-weighted ensemble: 2 BUY (0.57+0.75=1.32) > 1 SELL (0.62). Tech HEALTHY_TREND + sentiment flow reversal outweigh fundamental basket concern. +1 (not +3) due to dissent. SIZING: 50% target entry, add on 50DMA pullback or breakout.	AVERSE: SELL -2 (gap = 3!) Averse mode upweights bear language: Fund's ROIC<WACC + IRA dependence becomes primary driver. Subsidy concentration in 73% of basket = unacceptable risk under averse mandate. SIZING: Avoid wrapper. Wait for ROIC inflection OR 1Y sustained inflows.
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All LLM work inside Claude Code session (Plan-billed) — server never calls LLM

Per-ticker cost

Pattern	Sub-agent calls	Token usage	Wall time	Synthesis
CONSENSUS / ENTRY_TIMING (R1 stop)	3	~50k	~30-40s	rule-based (0)
POLAR_SPLIT / WIDE_SPAN (R2)	6	~95k	1-1.5 min	rule-based (0)

20-ticker Tier-A full cycle (assume 70% R1 only / 30% R2)

14 tickers x 3 calls (R1) = 42 calls
6 tickers x 6 calls (R1+R2) = 36 calls

TOTAL: ~78 sub-agent calls | ~5-7 min wall time | ~1.2M tokens
All billed to Claude Max plan — no external API spend.

Responsibility Boundary

IN-SESSION (Claude Code, Plan-billed)

Step 3: build prompts (specialist_prompts.py)
Step 4: ROUND 1 — 3 parallel sub-agent calls (Agent tool, market-researcher)
Step 5: parse free-text -> SpecialistOpinion
Step 7: ROUND 2 cross-exam (if needed)
Step 8: rule-based synthesis (Neutral/Averse)
Step 9a: cache write (.json)

Trigger:
- Manual: 'run today multi-agent debate'
- Scheduled: CronCreate (fresh session)

SERVER (uvicorn, LLM call ZERO)

Step 1: data refresh (price_discovery.py)
Step 2: SELECTION — BuyScore auto-compute (top_buy_picks / top_sell_picks)
Step 9b: /api/conviction-debate/multi
- selection live (fresh per request)
- cache lookup
- merge -> response
UI: React dashboard renders 4 sections

Trigger:
- Continuous (always-on uvicorn)
- Auto-refresh on /api/reload

Key Design Invariants

- LLM calls ONLY in-session. Server (uvicorn) NEVER calls an LLM. -> guarantees Plan-only spending.
- Selection re-computed every endpoint hit -> live scan immediately reflected in dashboard.
- Verdict cache immutable. New verdict overwrites only on next in-session run. -> reproducibility.
- Convergence-driven rounds. R2 only when needed -> cost discipline (~70% of tickers exit at R1).
- Dual synthesis (Neutral + Averse) -> single source serves all PM risk profiles.
- Lane enforcement -> prevents specialist domain creep; preserves intentional disagreement signal.